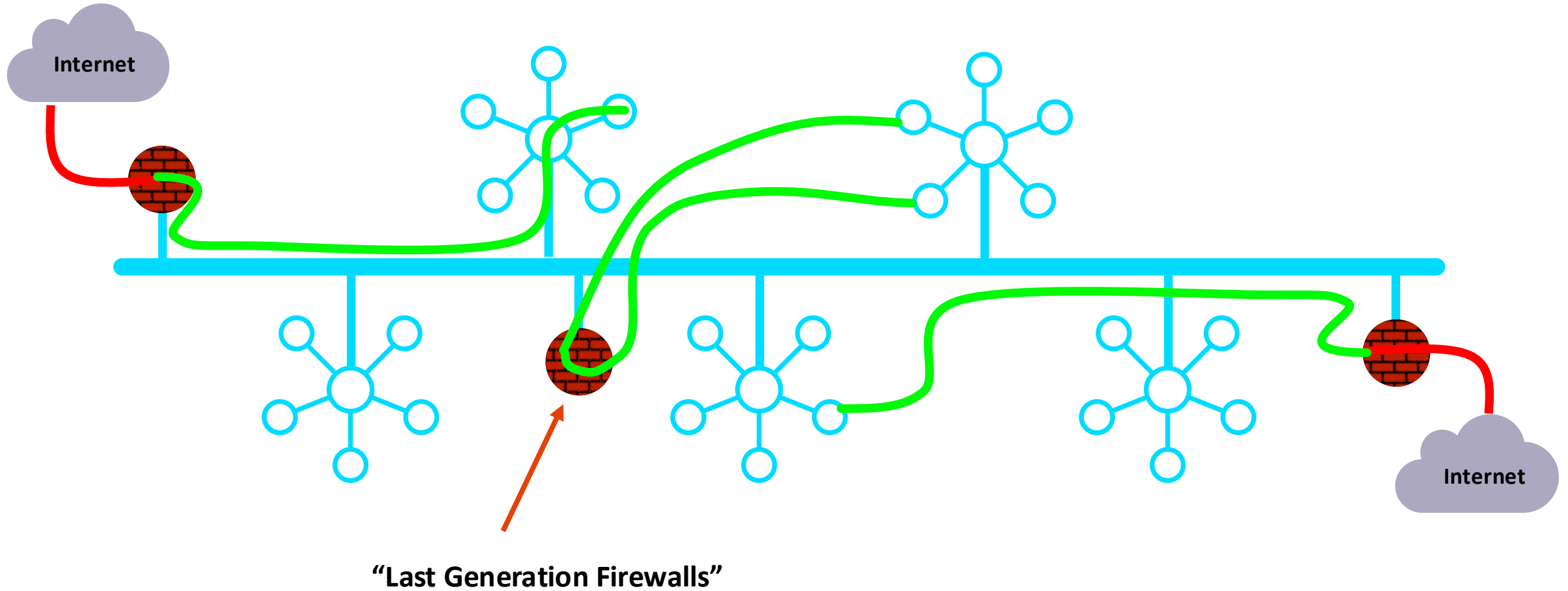




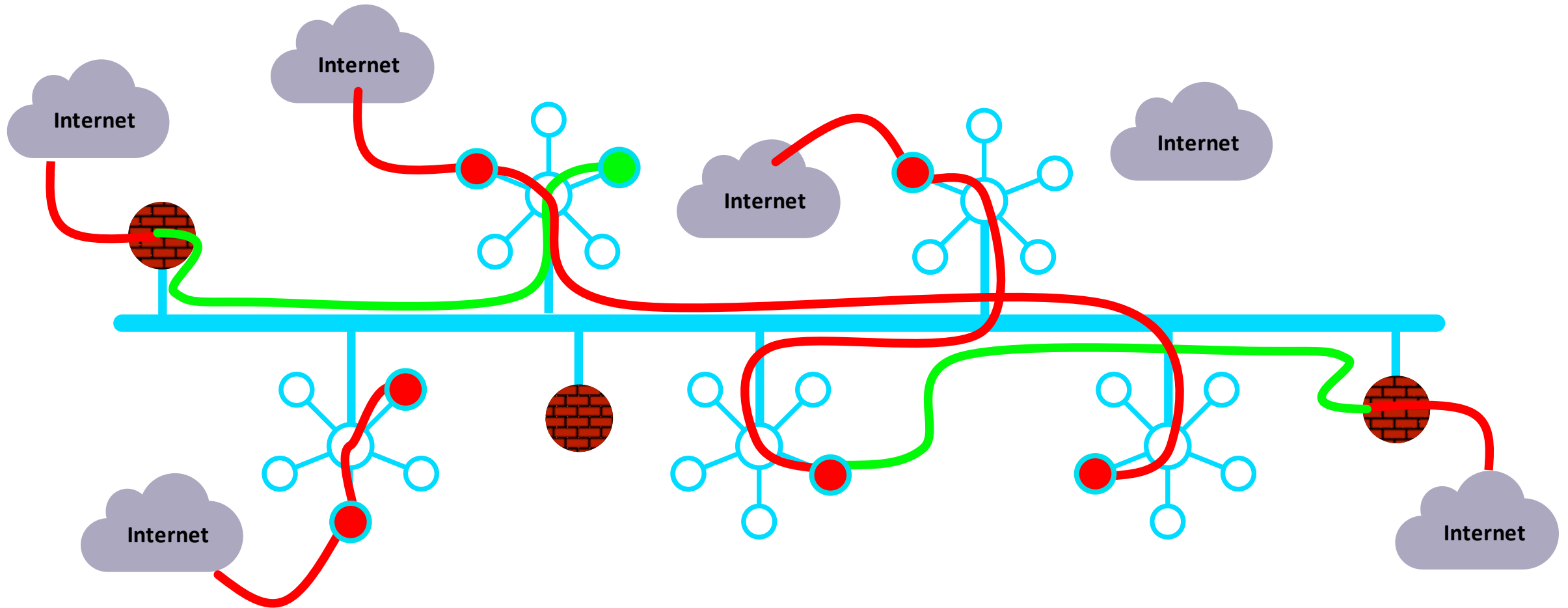
Distributed Cloud Firewall

ACE Team

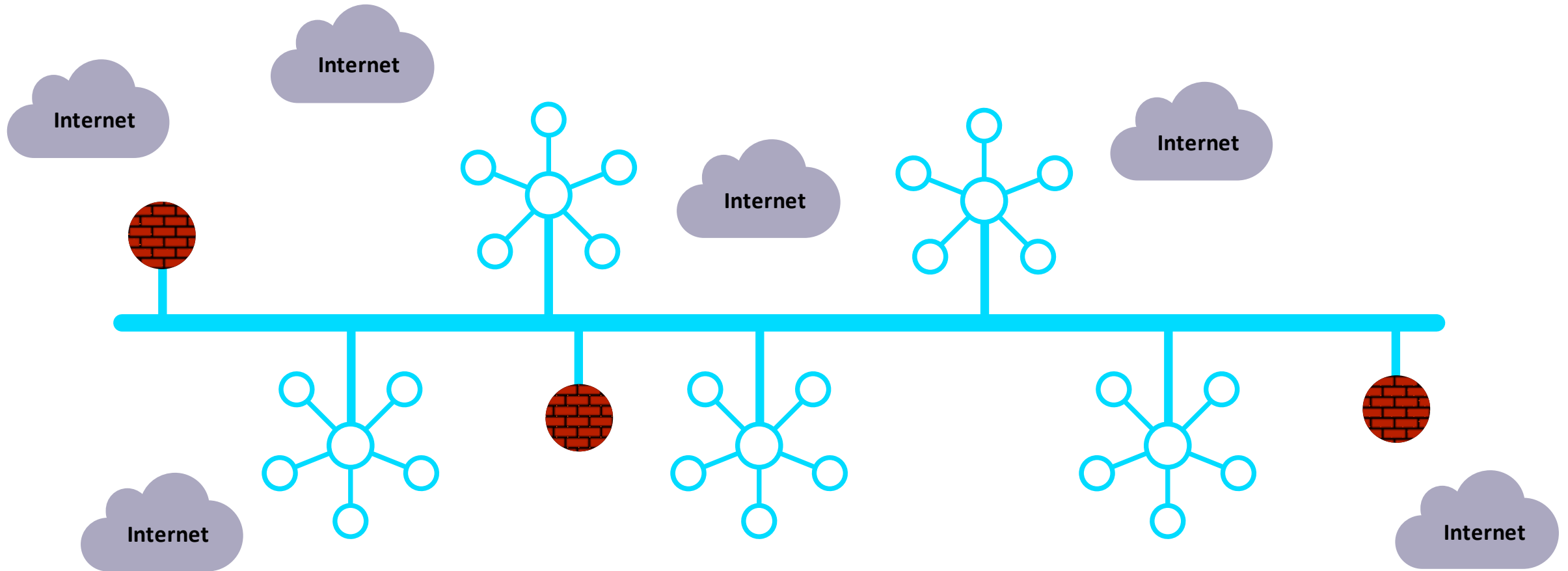
As Architected with Lift-and-Shift, Bolt-on, Data Center Era Products...



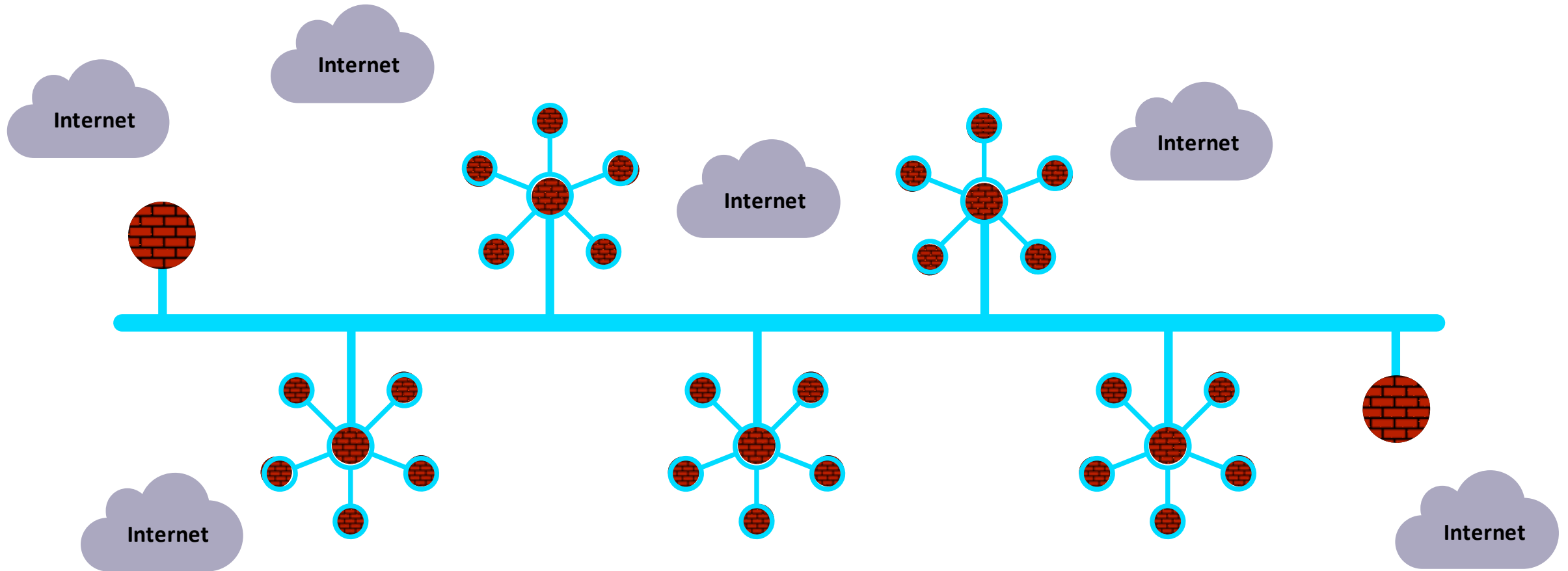
In Reality...



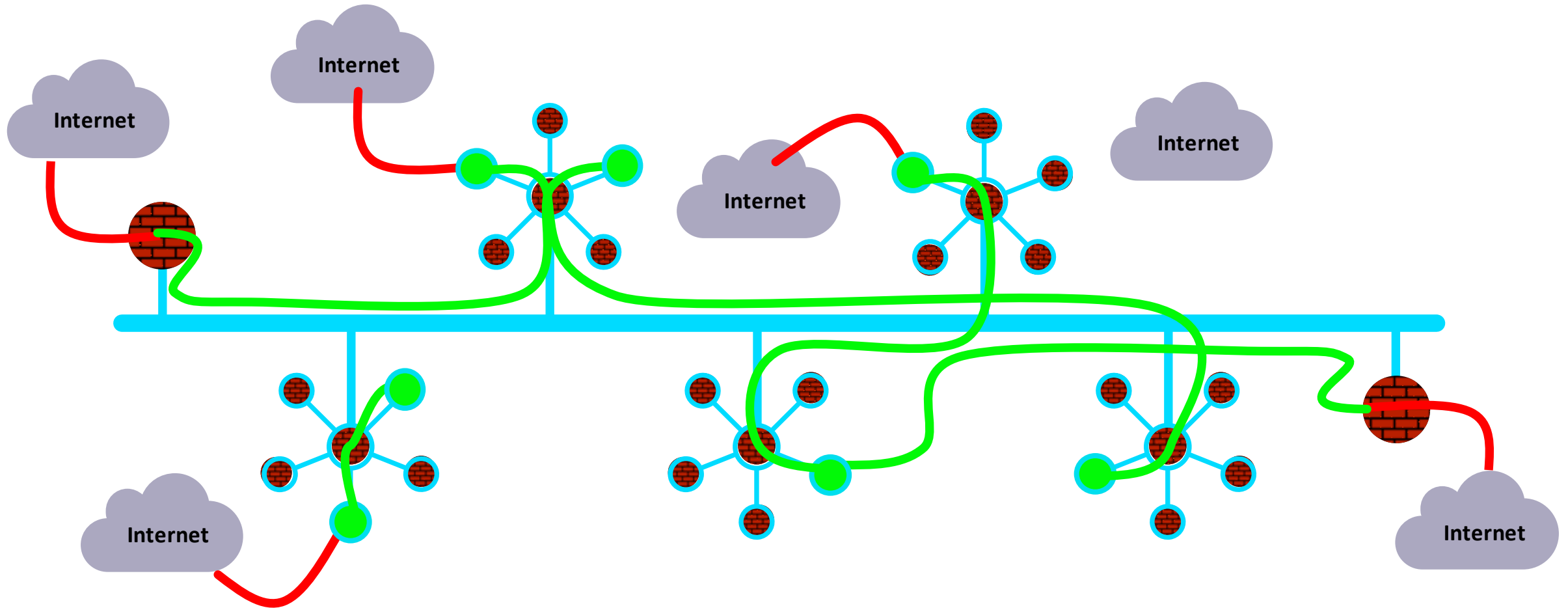
What If... the architecture was built for cloud



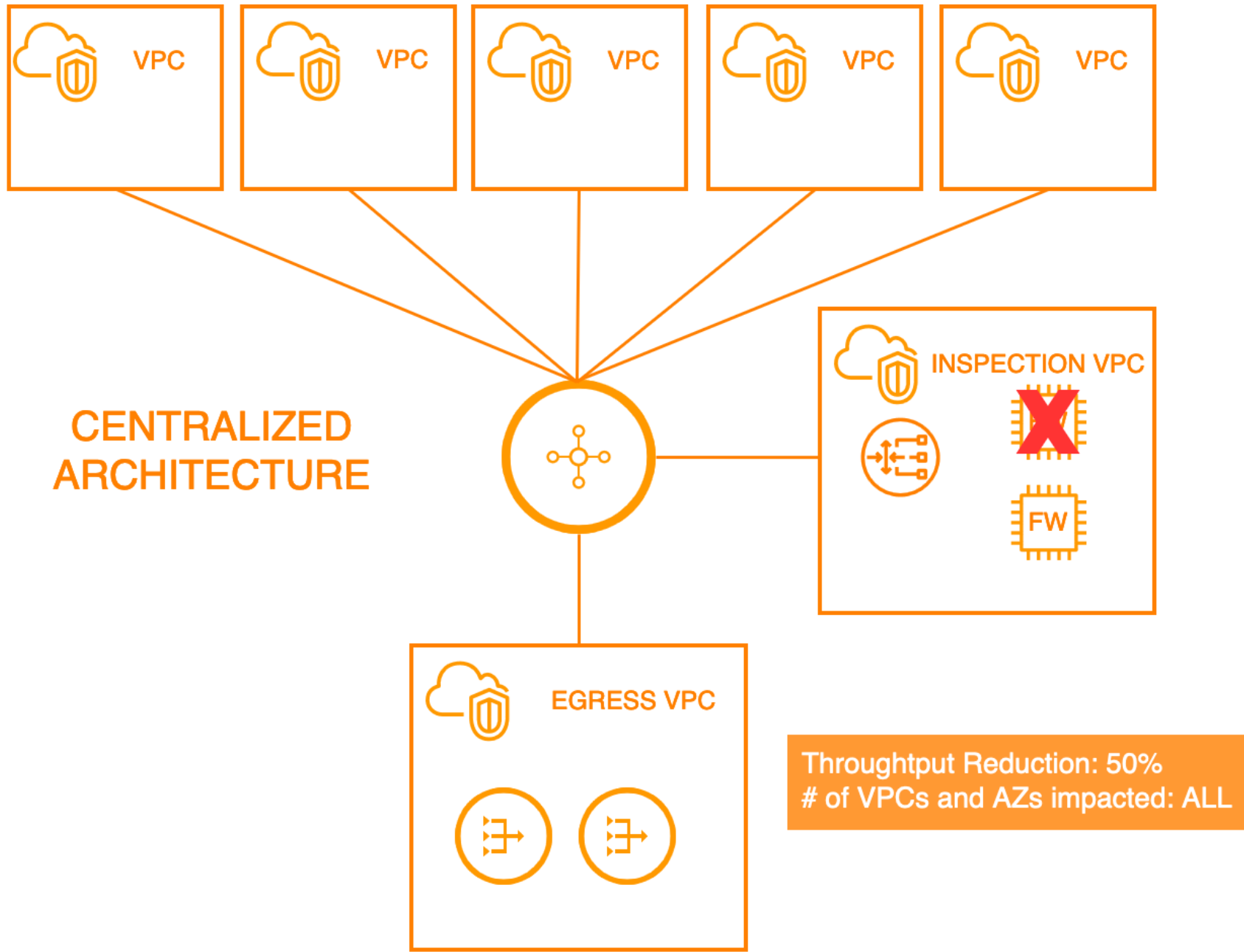
Firewalling Functions were Embedded in the Cloud Network Everywhere...



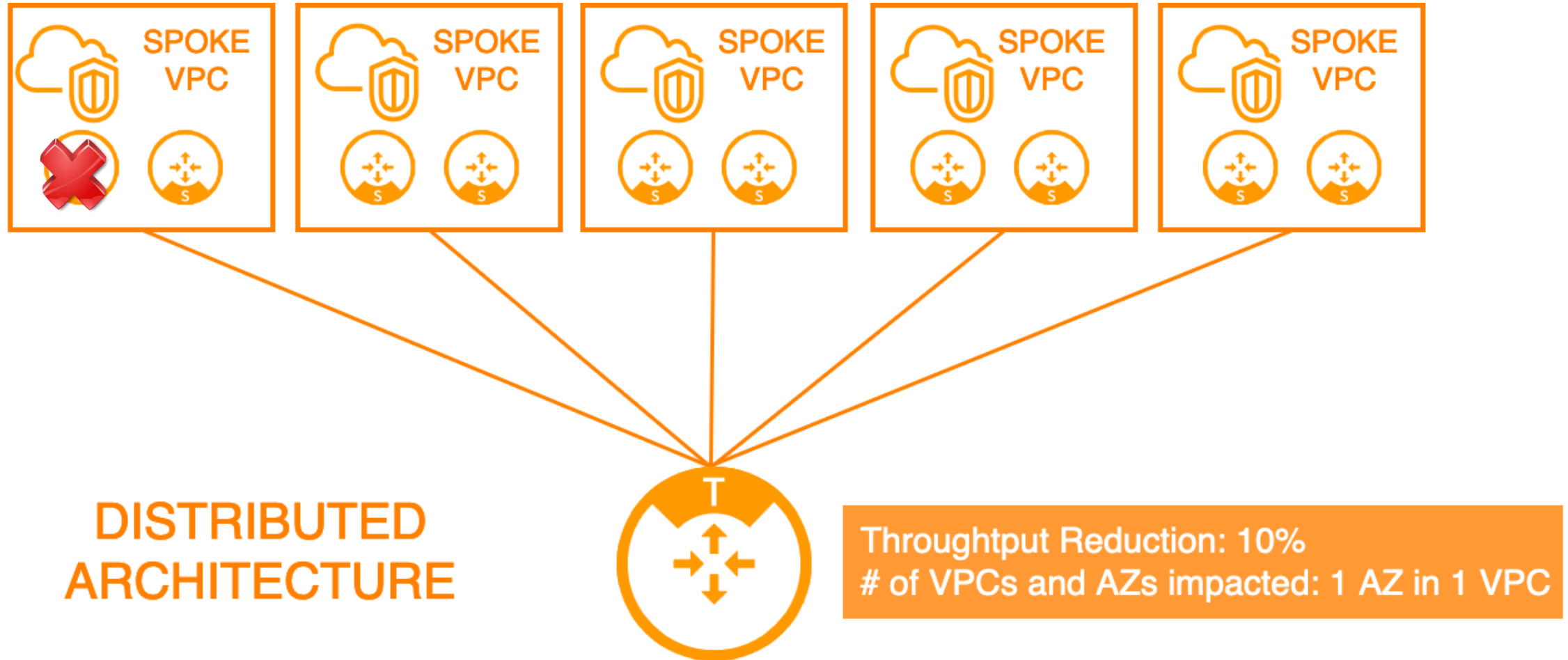
Distribution of the Security Services into the Spokes



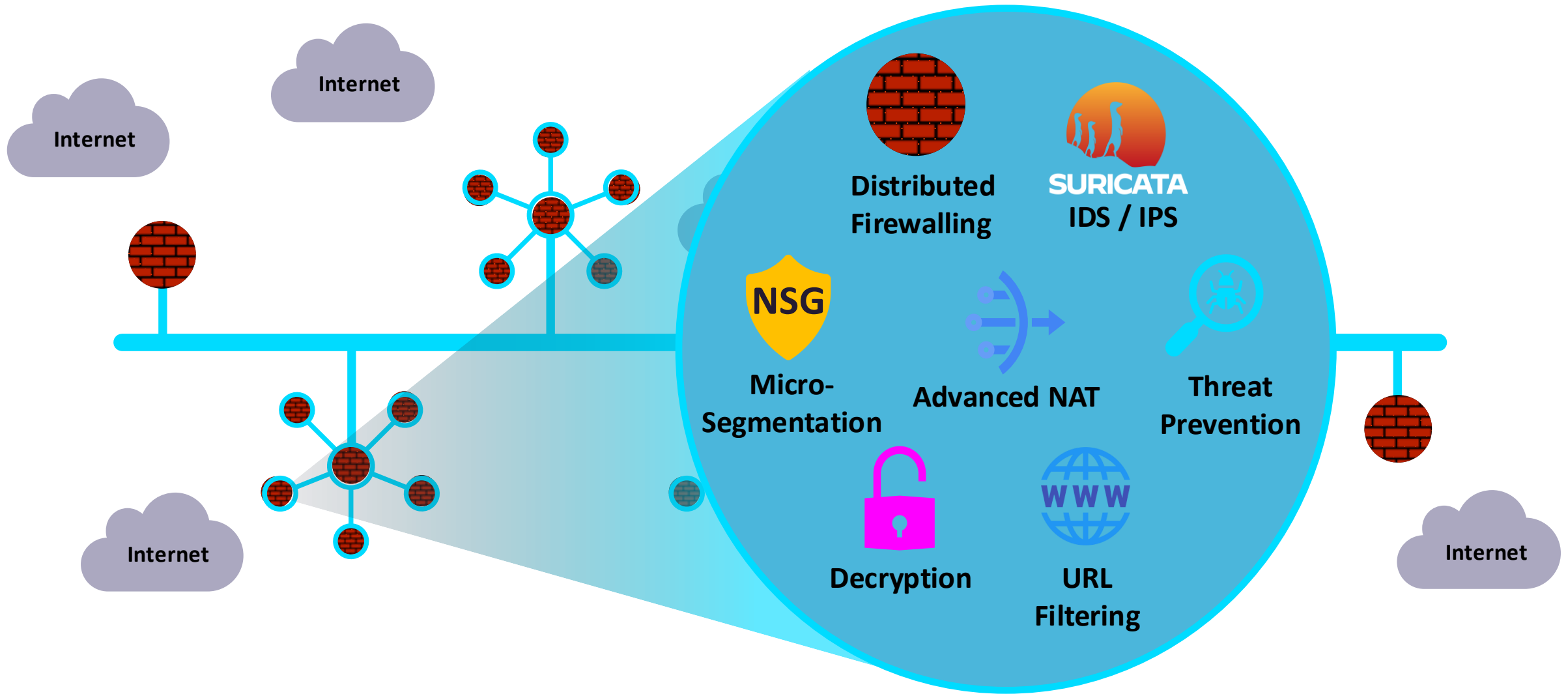
Impact of Failure – Centralized Architecture



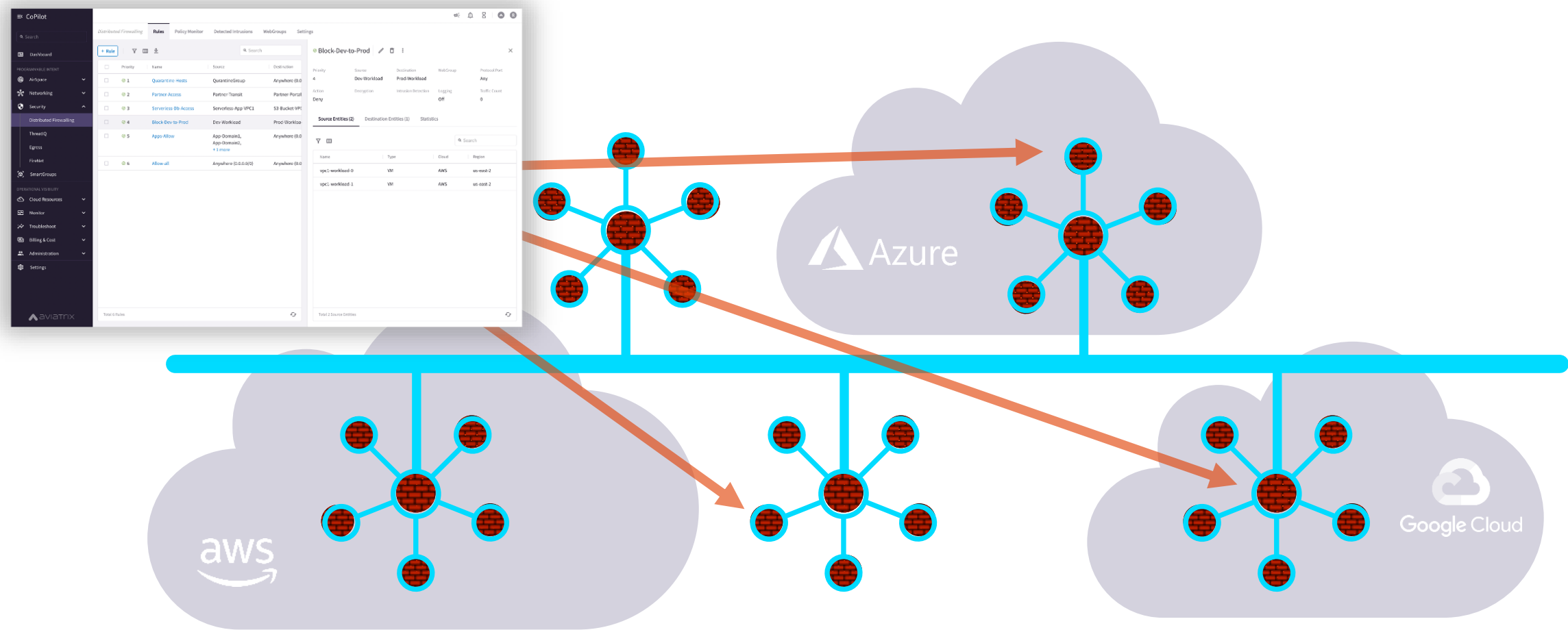
Impact of Failure – Distributed Architecture



And, What If it was more than just firewalling...



Policy Creation Looked Like One Big Firewall ... A Distributed Cloud Firewall...



Where and How Policies Are Enforced Is Abstracted...

SmartGroups: Definition

- A firewall rule consists of two important initial elements (i.e. *L3 info*):

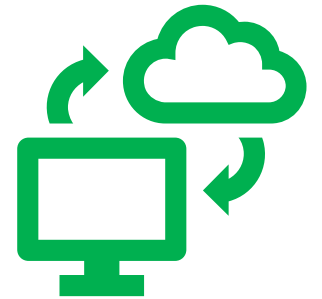
- ▣ Source
- ▣ Destination

- **What is a SmartGroup?**

A SmartGroup identifies a group of resources that have similar policy requirements and are associated to the same *logical container*.

- The members of a SmartGroup can be classified using *different* methods:

- CSP Tag
- Subnets
- VPC/Vnets
- Kubernetes
- Hostnames
- External Connections (S2C)



Smart Groups Creation



CoPilot

Groups: **SmartGroups** ExternalGroups WebGroups Settings

+ SmartGroup **Refetch CSP Resources**

Name: BU1

Resource Selection

Resource Types: VM, Subnet, and VPC/VNet are supported only on public AWS, Azure, and GCP clouds.

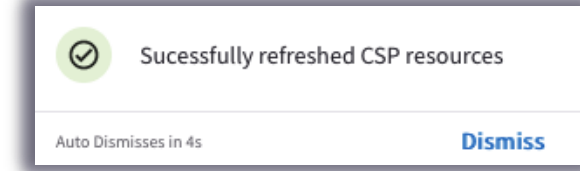
+ Resource Type

Virtual Machines

Matches all conditions (AND)

Environment dev

Cancel Save



Create SmartGroup

Name: BU1

Resource Selection Preview (3) ☒

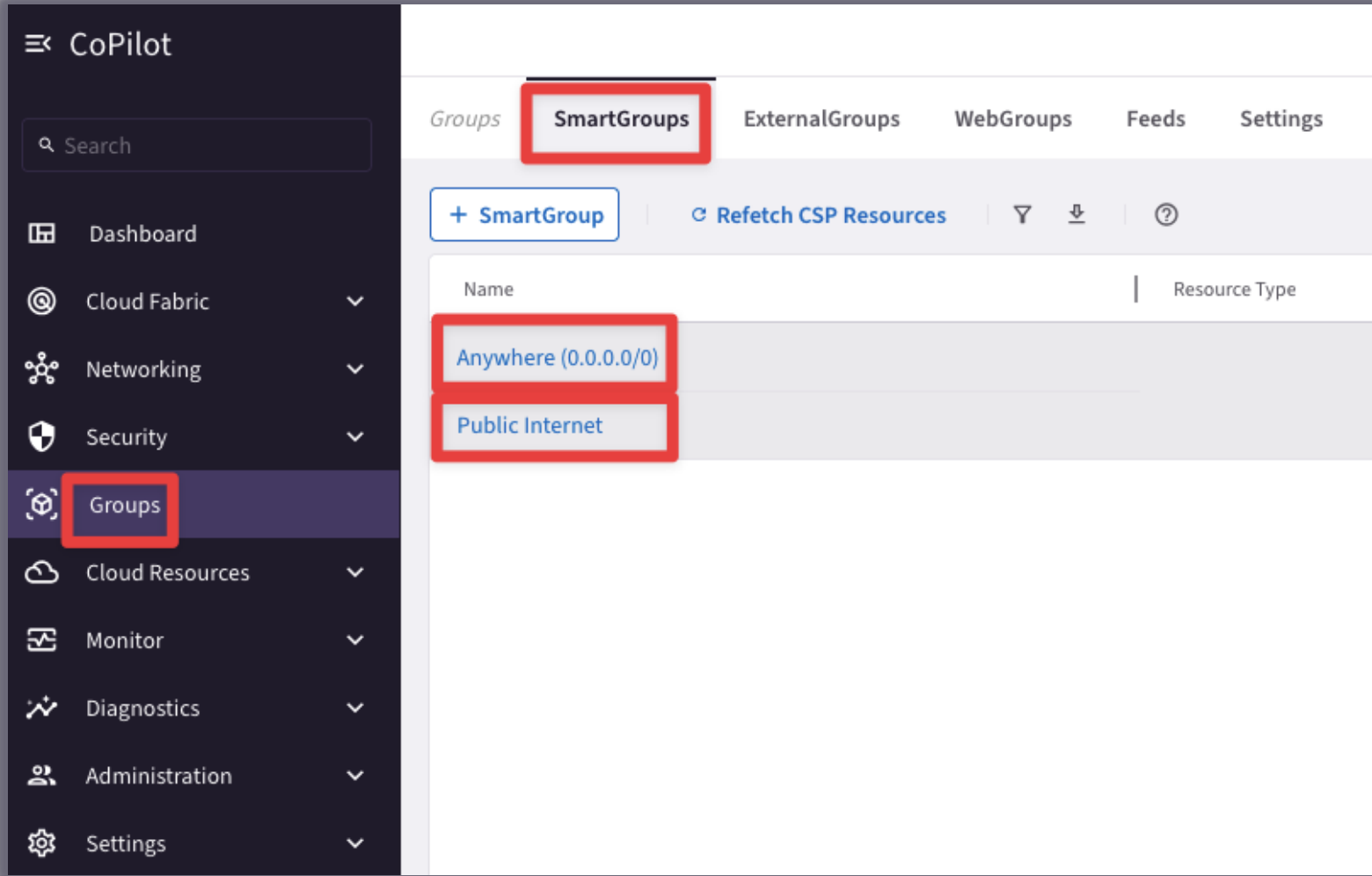
Name	Type	Cloud	Region
accounting-web-dev	VM	AWS	us-east-1
engineering-web-dev	VM	AWS	us-east-2
marketing-web-dev	VM	Azure ARM	northeurope

Total 3 Resources

Cancel Save

- Controller polls the CSPs to retrieve inventory (about VPCs, instances etc.) every **15 minutes** (can be modified)
- CoPilot queries Controller every **1 hour** (can be modified)
- On-demand refresh of tags is available

Pre-defined Smart Groups

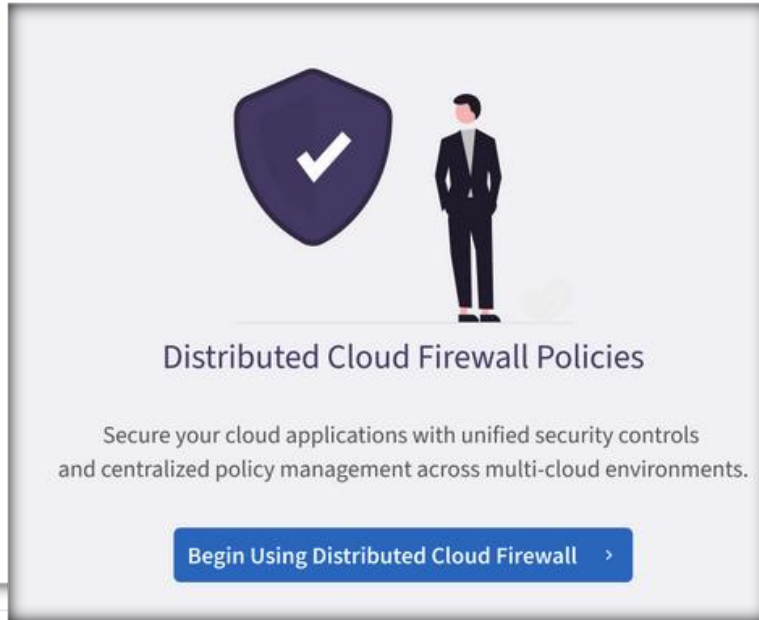


The screenshot shows the Aviatrix CoPilot interface. On the left is a dark sidebar with a 'CoPilot' header and a search bar. Below the search bar are menu items: Dashboard, Cloud Fabric, Networking, Security, Groups (highlighted with a red box), Cloud Resources, Monitor, Diagnostics, Administration, and Settings. The main panel has a top navigation bar with tabs: Groups, SmartGroups (highlighted with a red box), ExternalGroups, WebGroups, Feeds, and Settings. Below the tabs is a toolbar with a '+ SmartGroup' button, a 'Refetch CSP Resources' button, and icons for filter, download, and help. A table below the toolbar has two columns: 'Name' and 'Resource Type'. It contains two rows: 'Anywhere (0.0.0.0/0)' and 'Public Internet', both of which are highlighted with red boxes.

Name	Resource Type
Anywhere (0.0.0.0/0)	
Public Internet	

- **Anywhere (0.0.0.0/0)** → RFC1918 routes + Default Route (IGW)
- **Public Internet** → Default Route (IGW)

Enabling Distributed Cloud Firewall



- Distributed Cloud Firewall (DCF) uses micro-segmentation to provide granular network security rules for distributed applications in the Cloud. Distributed Cloud Firewall enables network policy enforcement between SmartGroups, WebGroups, and ExternalGroups you define in a single cloud or across multiple clouds.

Distributed Cloud Firewall | **Policies** | Monitor | Detected Intrusions | Settings

Ruleset: Post Rules Policy List (System) | [Manage Rulesets](#)

[+ Rule](#) | Actions |

Search | Modified View

☐ Priority	Name	Source	Destination	WebGroup	Protocol	Ports	Action
☐ 214748...	Default Action Rule	Anywhere (0.0.0.0/0)	Anywhere (0.0.0.0/0)		Any		Permit

Supported Cloud Providers and Gateways

Distributed Cloud Firewall is supported for the following clouds:

- AWS, AWS GovCloud, AWS China
- Azure, Azure Government, Azure in China
- GCP, Google for Government

The following gateway types are supported:

- Spokes attached to a Transit Gateway
- Spokes detached from a Transit Gateway
- Public Subnet Filtering Gateways ([enable PSF Gateways with DCF here](#))
- External connections (Site2Cloud) ([enable External Connections with DCF here](#)):
- Terminating on a Spoke Gateway
- Terminating on a Transit Gateway ([L4](#) only)
- [Edge as Spoke Gateway](#) ([L4 only](#); non-CSP tag)

How to create a Greenfield-Rule

Edit Rule: Greenfield-Rule

⚠ Rules will be applied only on AWS, AWS Gov, ARM, ARM Gov, and GCP

Name
Greenfield-Rule

Source SmartGroups
Anywhere (0.0.0.0/0) x v

Destination SmartGroups
Anywhere (0.0.0.0/0) x v

WebGroups
v

Protocol
Any v

Port
All
Specify multiple ports (e.g. 80) and/or port ranges (e.g. 80-8080)

Rule Behavior Enforcement ☒ Logging ☐

Action
Permit v

SG Orchestration [ⓘ]
☐ Off

Ensure TLS
☐ Off

TLS Decryption
☐ Off

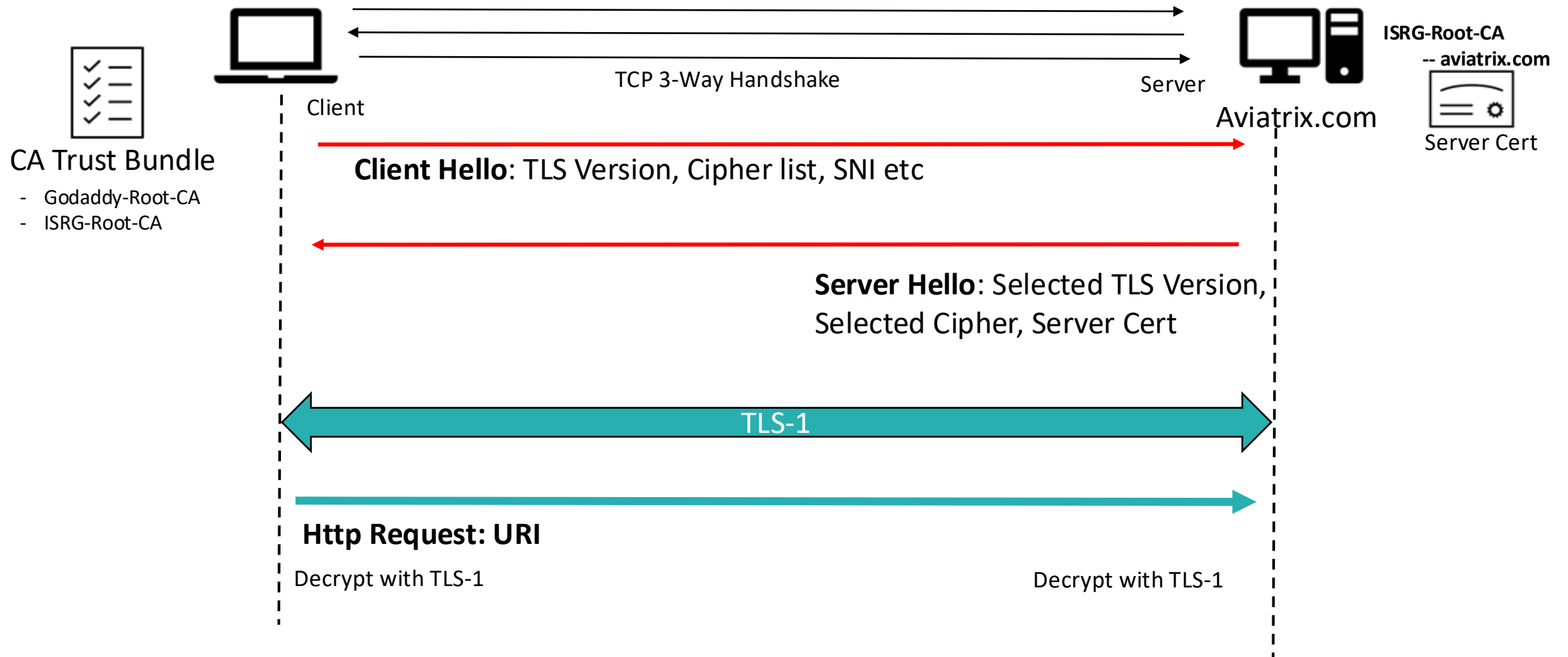
Intrusion Detection (IDS)
☐ Off

Rule Priority

Cancel Save In Drafts

- **Source SmartGroups:** Anywhere(0.0.0.0/0)
- **Destination SmartGroups:** Anywhere(0.0.0.0/0)
- **Protocol:** Any
- **Action:** Permit

TLS Decryption: Basic TLS Connection



TLS Decryption: PKI/ KMS and Trust Bundle

Certificate Hierarchy

- Root
 - Intermediate
 - Server Cert (Leaf Cert)

Certificate Fields

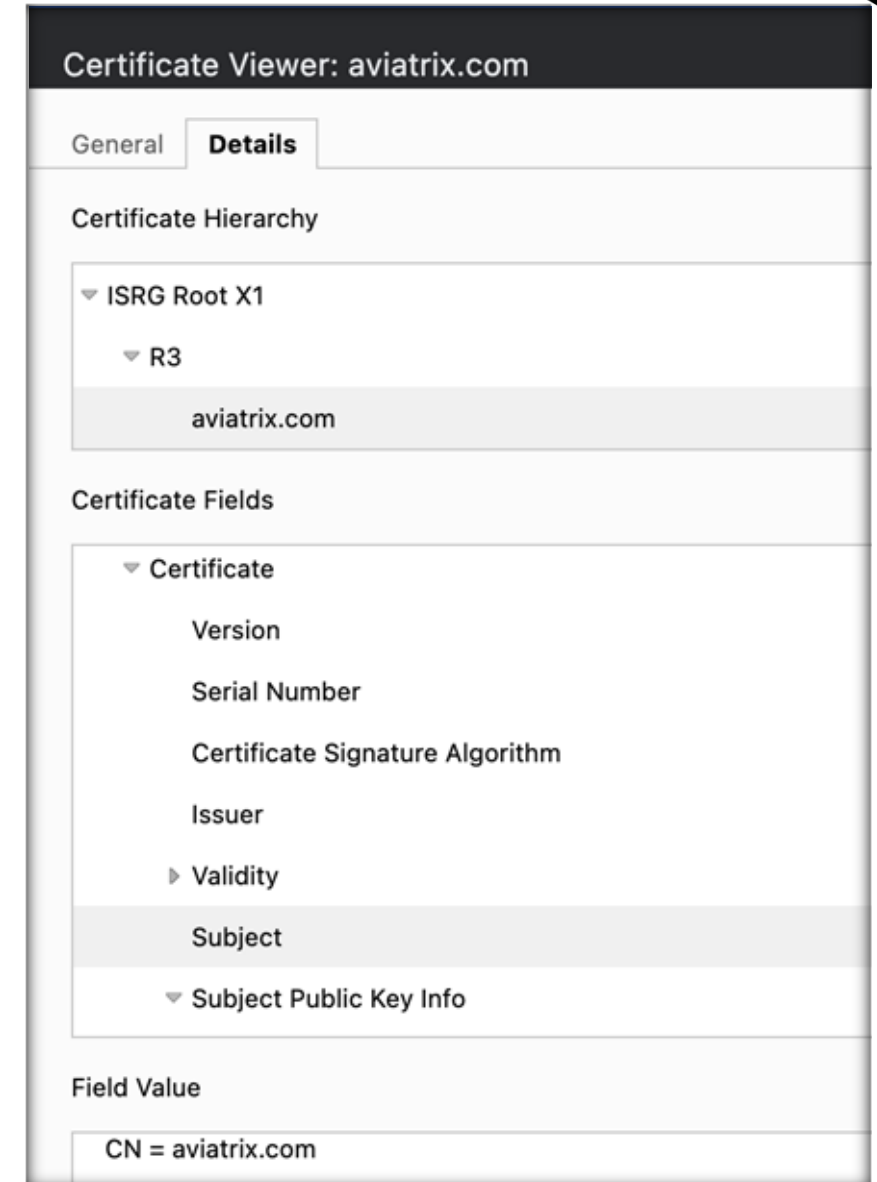
- Issuer
- Validity
- Subject

Trusted Root CA Bundle

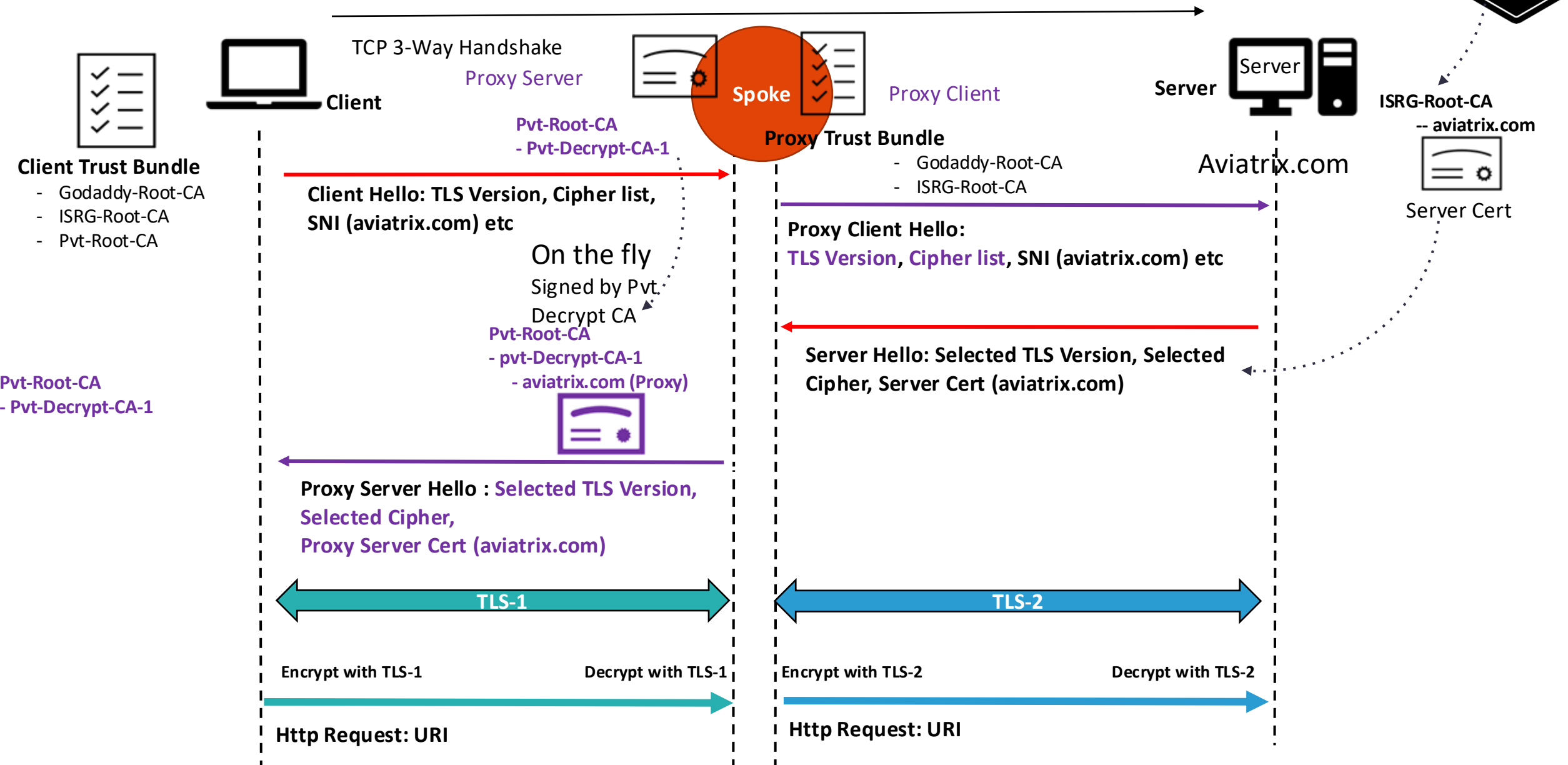
Used by the Client and/or Proxy Gateway to Identify/ Trust the Original Server Cert

Decryption CA Cert

Used by the Decryption/Proxy gateway to generate a new Proxy-Server Cert and Sign it with the Decryption CA Cert



TLS Decryption: Basic TLS Decryption



TLS Decryption: Decryption CA Cert

1. Download the Decryption CA Bundle.
2. Distribute the bundle across all the workloads.

❏ Decrypt CA Certificates should be trusted by the Source SmartGroup virtual machines when TLS Decryption is enabled for proxy.

Action

Permit

SG Orchestration ⓘ

On

Ensure TLS

Off

TLS Decryption

On

Intrusion Detection (IDS)

Off

Decrypt CA Certificates should be trusted by the **Source SmartGroup** virtual machines when TLS Decryption is enabled for proxy.

Distributed Cloud Firewall Rules Monitor Detected Intrusions Settings

Security Group (SG) Orchestration [Preview](#)

SG Orchestration adds control for both Intra-VPC Traffic and Inbound Internet Access on desired VPC/VNets.

Orchestration Enabled On

Complete 1 VPC/VNets

[Pause Next Cycle](#)

[View in Topology](#) [Manage](#)

Decryption CA Certificate

Certificate

Expires in 10 years (Self-Signed)

[Renew Certificate](#)

Enforcement

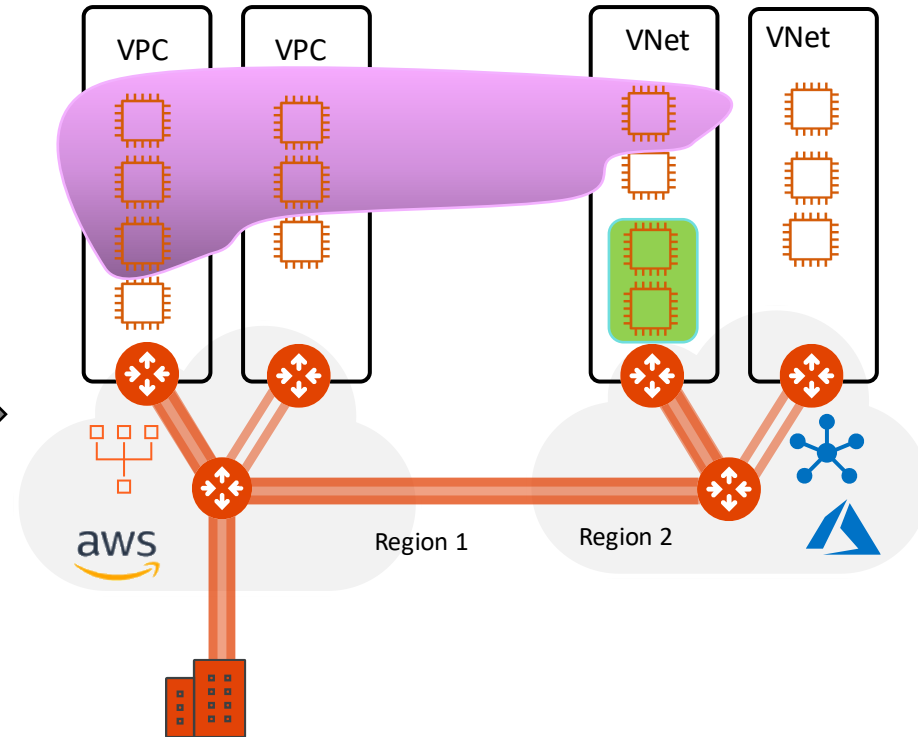
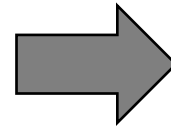
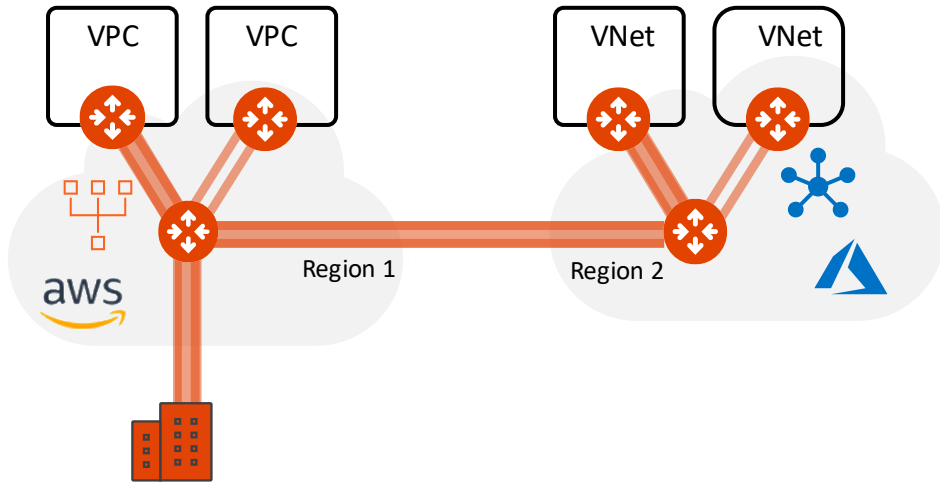
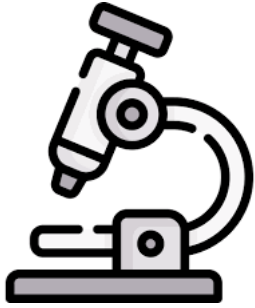
Permissive

Trust Bundle

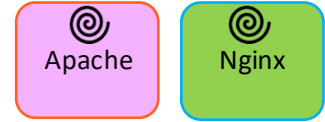
default-trustbundle

[Download Certificate](#)

Distributed Cloud Firewall Rule Types: Intra-rule vs. Inter-rule



Smart Groups

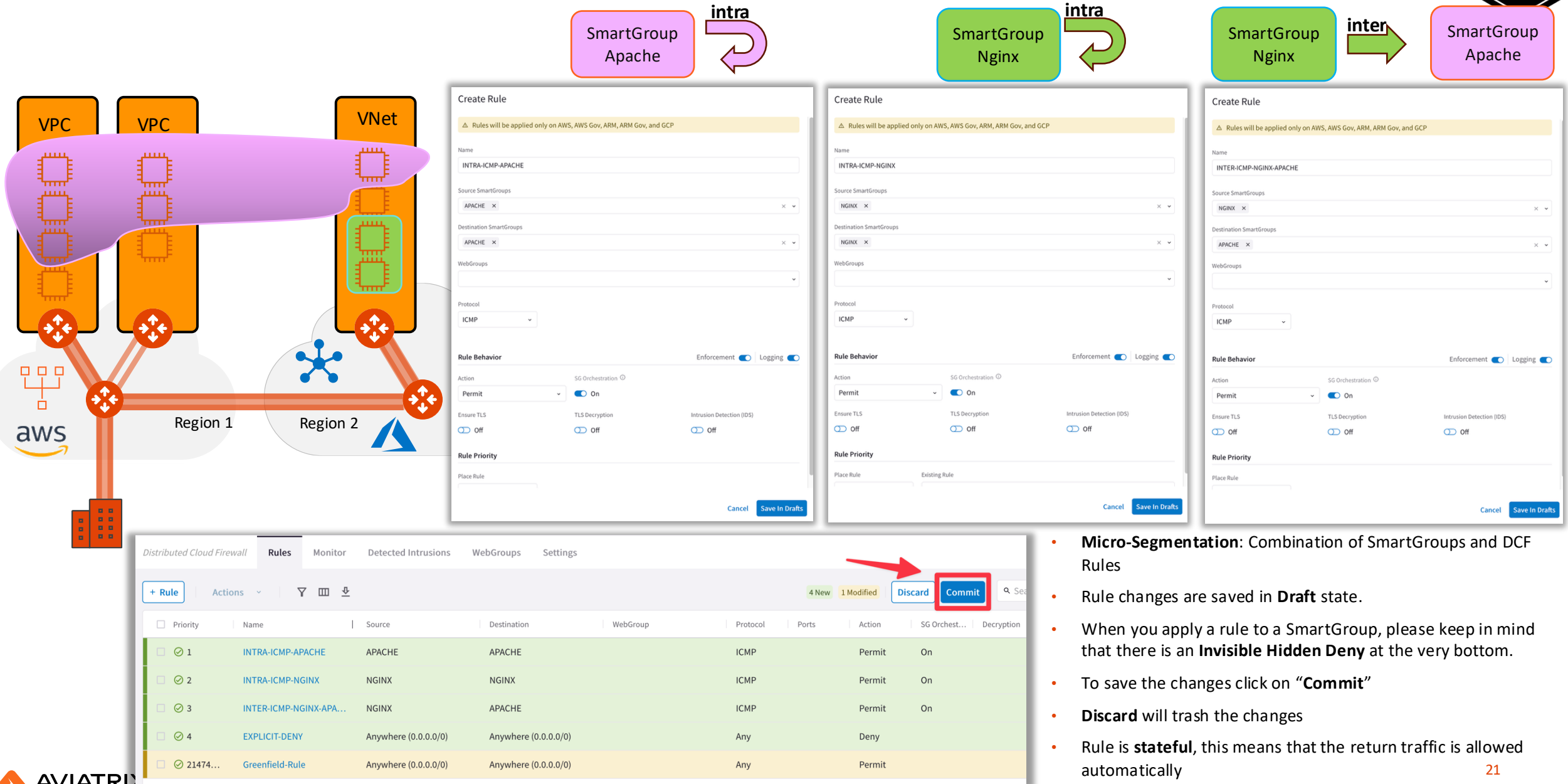


A rule between SGs can be defined for achieving the *INTER-SMARTGROUP* communication

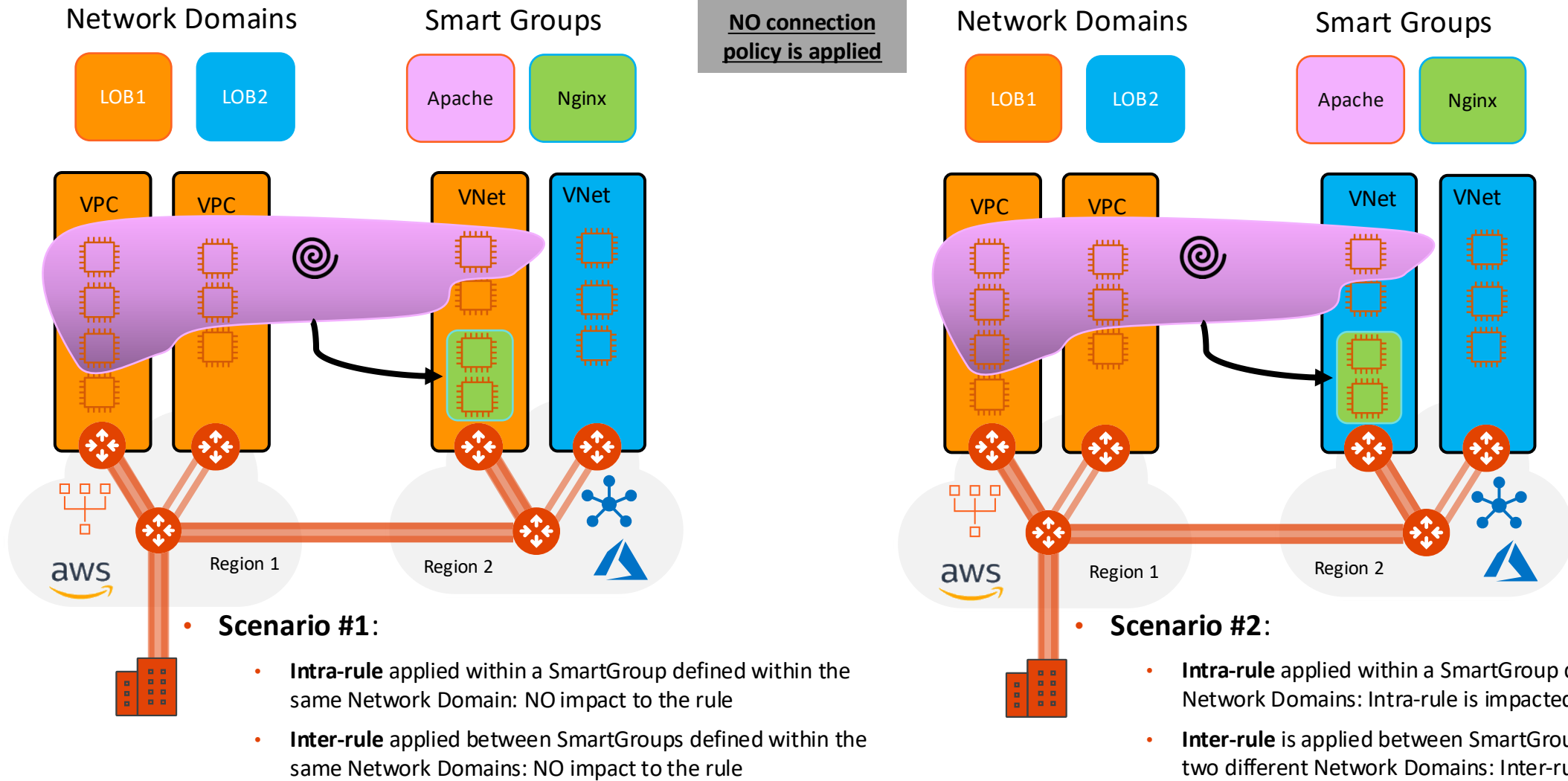
- **INTRA-RULE:** is defined within a Smart Group, for dictating what kind of traffic is allowed/prohibited among all the instances that belong to that Smart Group

- **INTER-RULE:** is defined among Smart Groups, for dictating what kind of traffic is allowed/prohibited among two or more Smart Groups.

Micro-Segmentation: SmartGroups, Intra-Rules and Inter-Rules



Network Segmentation & Distributed Cloud Firewall Rule together

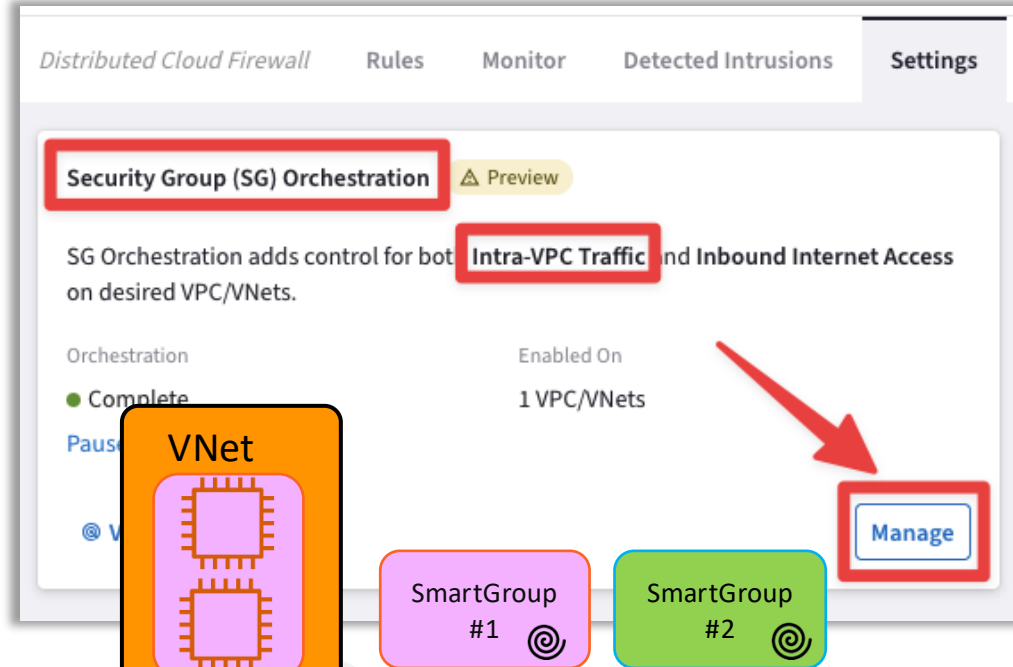


Caveat:

- Network Segmentation and Distributed Firewalling are **NOT** mutually exclusive!
- Network Segmentation takes **precedence** over the extent of a SmartGroup

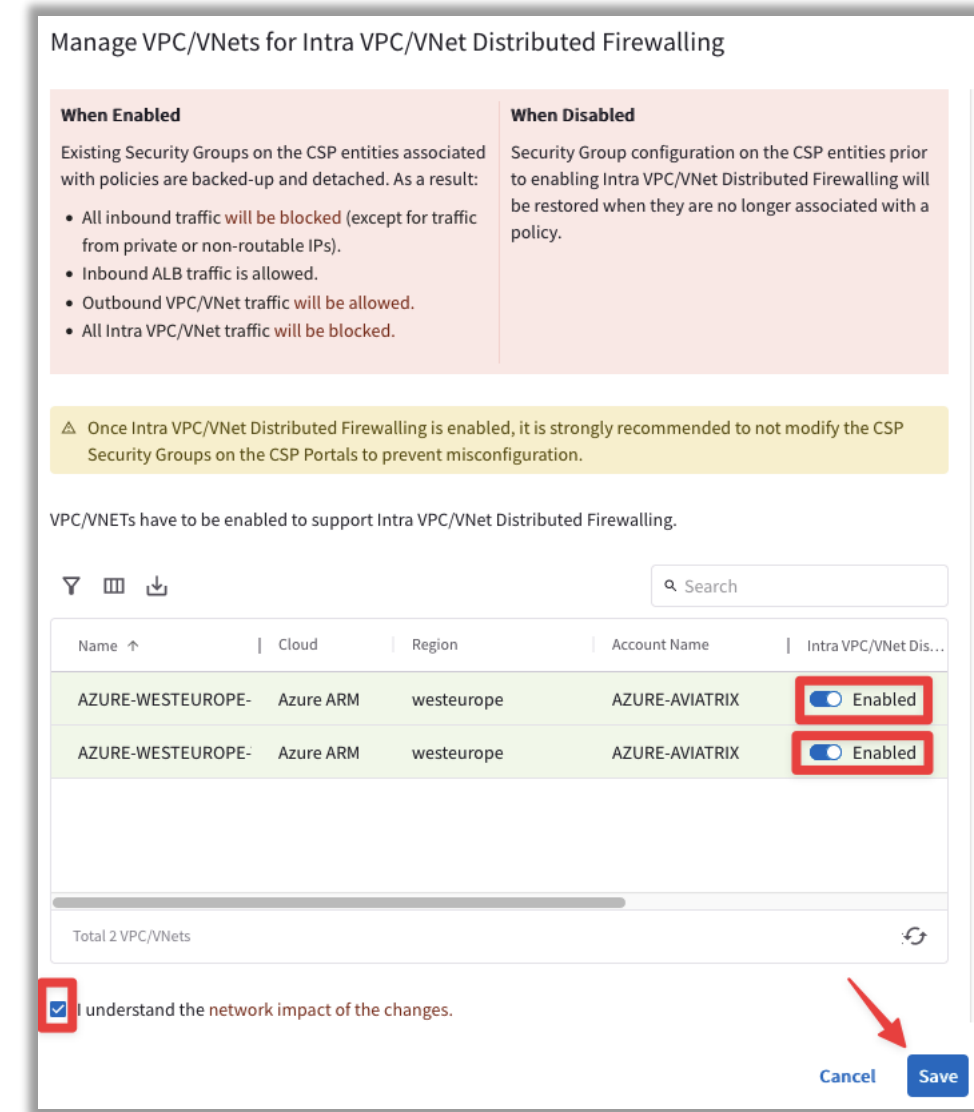
Security Group (SG) Orchestration: Intra VPC/VNET Traffic Control

□ Enable the feature on the relevant VPC/VNet



- If you enable the **Security Group (SG) Orchestration** (aka *Intra-VPC Traffic Control*), the SmartGroups defined within the same VPC/VNet will not be able to communicate with each other, unless an inter rule is applied between them.
- This is pure L4 separation, leveraging the Native Cloud Constructs (such as SG, NSG and ASG). This is not L7 inspection.

CAVEAT: Available in AWS/Azure



Rule Enforcement



Create Rule

⚠ Rules will be applied only on AWS, AWS Gov, ARM, ARM Gov, and GCP

Name
Allow-HTTPS

Source SmartGroups
AVX-FRANKFURT-PROD1

Destination SmartGroups
Public Internet

WebGroups
Any-Web

Protocol
TCP

Port
443

Specify multiple ports (e.g. 80) and/or port ranges (e.g. 80-8080)

Rule Behavior

Enforcement ☒ Logging ☐

Action
Permit

SG Orchestration ☐ Off

Ensure TLS ☐ Off

TLS Decryption ☐ Off

Intrusion Detection (IDS) ☐ Off

Rule Priority

Cancel Save In Drafts

☐ Enforcement ON

- Policy is enforced in the Data Plane

☐ Enforcement OFF

- Policy is NOT enforced in the Data Plane
- The option provides a *Watch/Test* mode
- Common use case is with deny rule
- Watch what traffic hits the deny rule before enforcing the rule in the Data Plane.

Rule Logging



Create Rule

Rules will be applied only on AWS, AWS Gov, ARM, ARM Gov, and GCP

Name: Allow-HTTPS

Source SmartGroups: AVX-FRANKFURT-PROD1

Destination SmartGroups: Public Internet

WebGroups: Any-Web

Protocol: TCP, Port: 443

Specify multiple ports (e.g. 80) and/or port ranges (e.g. 80-8080)

Rule Behavior

Action: Permit

SG Orchestration: Off

Ensure TLS: Off

TLS Decryption: Off

Intrusion Detection (IDS): Off

Rule Priority

Cancel Save In Drafts

Distributed Cloud Firewall

Rules Monitor Detected Intrusions Settings

Auto Refresh

Search

All Logs

Timestamp	Rule	Source IP	Destination IP	URL	Protocol	Source Port	Destination Port	Action	Enforced
Mar 25, 2025 5:54:04 PM	default-deny-all	10.2.5.141	10.4.2.10		TCP	44324	3306	Deny	On
Mar 25, 2025 5:54:03 PM	default-deny-all	10.2.5.149	10.4.2.10		TCP	57200	3306	Deny	On
Mar 25, 2025 5:54:03 PM	allow-internet-https	10.2.2.40	209.85.202.138		TCP	56834	443	Permit	On
Mar 25, 2025 5:54:03 PM	allow-internet-https	10.2.2.40	23.217.72.114		TCP	44650	443	Permit	On
Mar 25, 2025 5:54:03 PM	allow-internet-https	10.2.2.70	209.85.203.102		TCP	57610	443	Permit	On
Mar 25, 2025 5:54:03 PM	default-deny-all	10.1.5.13	10.2.5.163		TCP	56230	443	Deny	On
Mar 25, 2025 5:54:03 PM	allow-internet-https	10.2.2.70	2.18.237.177		TCP	41148	443	Permit	On
Mar 25, 2025 5:54:01 PM	allow-k8s-prod-marketing	10.1.5.57	10.2.5.161		TCP	34700	443	Permit	On
Mar 25, 2025 5:54:01 PM	allow-internet-https	10.1.5.13	151.101.3.52		TCP	47030	443	Permit	On
Mar 25, 2025 5:54:01 PM	allow-internet-https	10.1.5.47	147.75.40.148		TCP	60574	443	Permit	On

Logging can be turned ON/OFF per rule

Configure Syslog to view the logs



DCF IPS

A server room with rows of racks and a glowing fiber optic device in the foreground.

What Is an Intrusion Prevention System (IPS)?

IPS inspects traffic **inline** and actively blocks attacks before they reach workloads. Unlike IDS which only generates alerts, IPS sits in-path and can drop packets, reset connections, or allow with alert.

Prevent Intrusions

Blocks attacks and lateral movement before they reach critical workloads

Stronger Compliance

Provides proof of "Zero Trust" and regulatory compliance requirements

Lower Operational Load

Reduces burden on SOC/IR teams through automated prevention

How Aviatrix Suricata-Based IPS Works

Aviatrix embeds Suricata inline in the Cloud Native Security Fabric for real-time, granular enforcement across all your cloud environments.

01

Inline Suricata via DCF

Suricata IPS engine embeds directly into the Cloud Native Security Fabric (CNSF), combining IPS with cloud-native performance and centralized policy control. Traffic is sent to Suricata engine for deep inspection and DCF enforces policy decisions in real-time—all within microseconds.

02

Intrusion Profiles = Control & Customization

Profiles map Suricata signature sets to per-severity actions (alert, alert+drop). Allow customers to ignore noisy signatures and define "never drop" rules for critical business flows (by SID). Use System defaults and apply custom rule sets per VPC for environment-specific tuning.

03

Operational Benefits

The feature is embedded into the DCF policy and it can be activated either through UI or driven by terraform/API

AWS and Azure (premium firewall) offer IPS features, but only within their own cloud, with fragmented policy and limited flexibility. Each platform requires separate configuration and management.

Clear Customer Value in the Multi-Cloud Market

One IPS Fabric. All Your Clouds.



Reduce Business Risk

Stop exploits and data exfiltration before they impact critical applications. Block advanced persistent threats and ransomware, protecting cloud workloads that span traditional VM to serverless applications.



Simplify Operations

One IPS policy model, one unified dashboard, one automation surface. Eliminate the complexity of managing separate security stacks for each cloud provider and reduce tool sprawl that drives up costs.



Accelerate Zero Trust

Combine network segmentation, workload controls, and real-time IPS enforcement in a single fabric. Meet compliance requirements with audit-ready evidence of consistent prevention controls.

The Challenge

Enterprises face fragmented security stacks per cloud and per business unit, making it nearly impossible to prove consistent controls to regulators and leadership.

The Impact

Multi-cloud and multi-team environments struggle with tool sprawl, driving up complexity and costs while leaving dangerous security gaps between cloud boundaries.

The Solution

Aviatrix delivers your **multi-cloud security fabric**, with Suricata-based IPS as a core control—standardizing policy and reporting from day one.

 **Board-Level Outcomes:** Reduced breach risk, simplified operations, and audit-ready evidence—all delivered through a single platform that scales with cloud growth



Architecture & Component Flow



1. IPS section

Distributed Cloud Firewall Policies Monitor Audit **IPS** TLS Settings

Profiles Rule Feeds

+ Custom Profile |

Name	Drop Level	Log Level
System Default IPS Profile Active	No Drops	All Severities

CSP	minimum GW-size
AWS	t3.medium
Azure	Standard_B2ms
GCP	n1-standard-1
OCI	VM.Standard2.2



Architecture & Component Flow



2. Rule Feeds (for custom Signature Rules-set)

The screenshot shows the 'Distributed Cloud Firewall' interface. The 'IPS' tab is selected in the top navigation bar. In the left sidebar, the 'Rule Feeds' tab is selected, and the '+ Custom Rule Feed' button is highlighted with a red box. A red arrow points from this button to the 'Add Custom Rule Feed' dialog box. The dialog box contains the following fields and buttons:

- Add Custom Rule Feed** (Title)
- Follow the rule format and include essential information.** (Instruction) with a [View Rule Format](#) link.
- Name** (Field):
- Signature File** (Field):
- Upload** (Button) - highlighted with a red box.
- Cancel** (Button)
- Upload** (Button)

Feed Types:

- External feeds (e.g., Suricata rules from Proofpoint)
- Custom customer-uploaded rules
- **Limitation:** One external feed per IPS profile (avoids conflicts, simplifies management)



Architecture & Component Flow



3. Custom Profile

Distributed Cloud Firewall Policies Monitor Audit IPS TLS Settings

Profiles Rule Feeds

+ Custom Profile

Name	Drop Level	Log Level
System Default IPS Profile	No Drops	All Severities

Create Custom Profile

Name: Custom-Profile

Rule Feed: ET Open

Drop Level: Only Critical Severity

Alert Level: All Severities

Cancel Save

Default Profile:

- The **System Default IPS Profile** is the default Suricata signature profile that contains the Suricata-recommended rule set.



Architecture & Component Flow



3. Custom Profile

The screenshot shows the 'Distributed Cloud Firewall' interface with tabs for Policies, Monitor, Audit, IPS, TLS, and Settings. The 'Profiles' tab is active, and a '+ Custom Profile' button is highlighted with a red box. Below this, a table lists the 'System Default IPS Profile' as 'Active' with 'No Drops' and 'All Severities'. A 'Create Custom Profile' dialog box is open in the center, with the following fields:

- Name:** 'Custom-Profile' (highlighted with a red box)
- Rule Feed:** 'ET Open' (highlighted with a red box)
- Drop Level:** 'Only Critical Severity' (dropdown menu)
- Alert Level:** 'All Severities' (dropdown menu)

At the bottom of the dialog are 'Cancel' and 'Save' buttons.

Default Profile:

- The **System Default IPS Profile** is the default Suricata signature profile that contains the Suricata-recommended rule set.

- Once you have created the Profile, it is enabled globally across the whole CNSF.



Architecture & Component Flow



4. DCF Policy and IPS

Default Profile:

- Turn on *Intrusion Analysis* and *TLS decryption* (for DPI) in **DCF policy**.

Create Rule

Name

IPS-Rule

Source Groups

WorkstationEdge

Destination Groups

aws-us-east-2-spoke1-test1

WebGroups

Protocol

Any

Port

All

Specify multiple ports (e.g. 80) and/or port ranges (e.g. 80-8080)

Rule Behavior

Enforcement On

Log At Start

Decrypt CA Certificates should be trusted by the Source SmartGroup virtual machines when TLS Decryption is enabled for proxy.

Action

Permit

SG Orchestration

Off

Ensure TLS

Off

TLS Decryption

On

Intrusion Analysis

On

TLS Profile

Rule Priority

Cancel

Save In Drafts



Architecture & Component Flow



5. Intrusion Logs

Distributed Cloud Firewall

Policies

Monitor

Audit

IPS

TLS

Settings

Policy Logs

Intrusion Logs

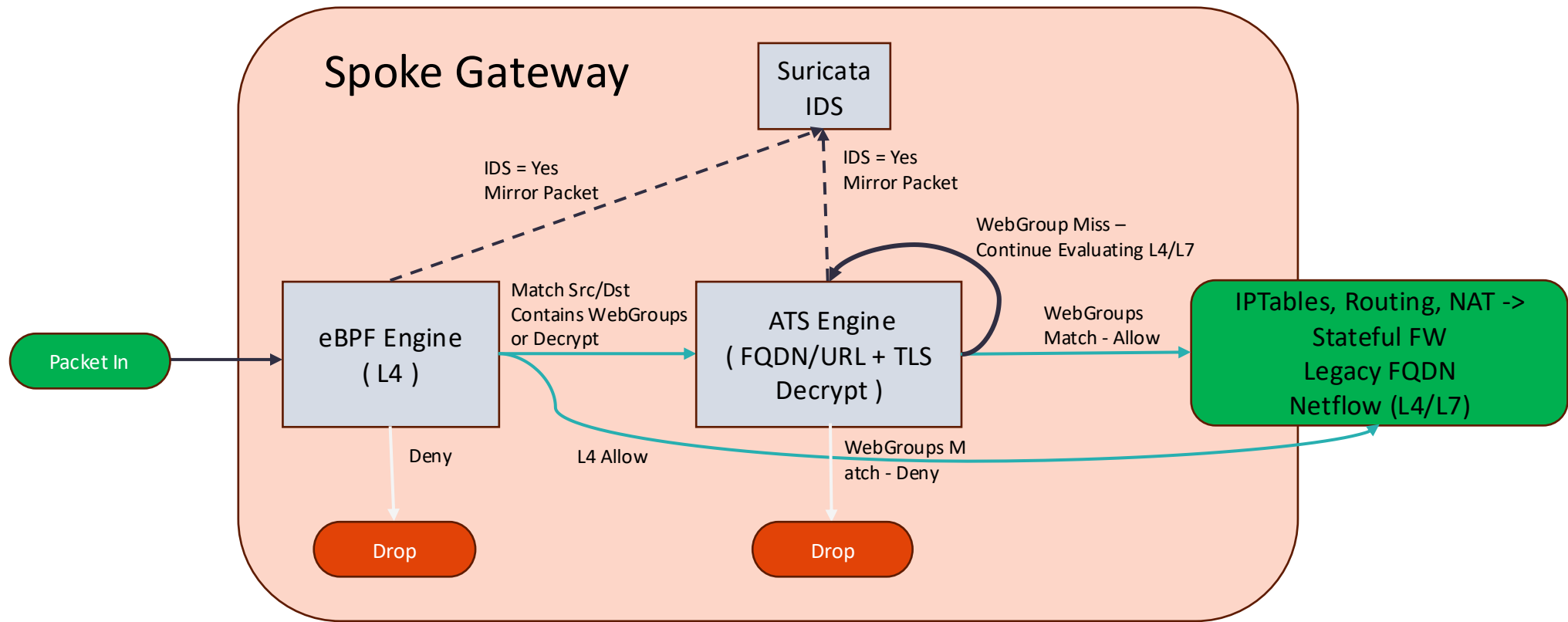
Search

Default View

Timestamp	Rule	SID	Severity	Source IP	Destination IP	Protocol	Source Port	Destination Port	Application Proto	Gateway	Category	Attack Target	Deployment
Jul 22, 2026 5:46 PM	allow-internet...	2013504	Information	10.2.2.40	51.104.174...	TCP	46702	80	http	GW-marke...	Not Suspic...		
Jul 22, 2026 5:43 PM	allow-internet...	2013504	Information	10.2.2.40	51.104.174...	TCP	42372	80	http	GW-marke...	Not Suspic...		
Jul 22, 2026 2:36 PM	allow-internet...	2013504	Information	10.4.3.10	35.203.149...	TCP	57736	80	http	GW-enterp...	Not Suspic...		
Jul 22, 2026 2:35 PM	allow-internet...	2013504	Information	10.4.3.10	104.198.11...	TCP	34172	80	http	GW-enterp...	Not Suspic...		
Jul 22, 2026 2:35 PM	allow-internet...	2013504	Information	10.4.3.10	91.189.91.81	TCP	59862	80	http	GW-enterp...	Not Suspic...		
Jul 22, 2026 2:07 PM	allow-internet...	2013504	Information	10.4.2.10	34.105.40.82	TCP	49358	80	http	GW-enterp...	Not Suspic...		
Jul 22, 2026 2:07 PM	allow-internet...	2013504	Information	10.4.2.10	34.169.122...	TCP	40492	80	http	GW-enterp...	Not Suspic...		
Jul 22, 2026 2:07 PM	allow-internet...	2013504	Information	10.4.2.10	91.189.92.22	TCP	52982	80	http	GW-enterp...	Not Suspic...		
Jul 22, 2026 1:28 PM	allow-internet...	2013504	Information	10.4.4.10	34.169.122...	TCP	55622	80	http	GW-enterp...	Not Suspic...		
Jul 22, 2026 1:27 PM	allow-internet...	2013504	Information	10.4.4.10	34.105.40.82	TCP	47114	80	http	GW-enterp...	Not Suspic...		
Jul 22, 2026 1:27 PM	allow-internet...	2013504	Information	10.4.4.10	185.125.19...	TCP	50510	80	http	GW-enterp...	Not Suspic...		
Jul 22, 2026 11:28 AM	allow-internet...	2013504	Information	10.2.2.45	51.104.174...	TCP	39116	80	http	GW-marke...	Not Suspic...		
Jul 22, 2026 11:28 AM	allow-internet...	2013504	Information	10.2.2.45	51.104.174...	TCP	36628	80	http	GW-marke...	Not Suspic...		
Jul 22, 2026 8:54 AM	allow-internet...	2013504	Information	10.2.2.100	51.104.174...	TCP	36194	80	http	GW-marke...	Not Suspic...		
Jul 22, 2026 8:54 AM	allow-internet...	2013504	Information	10.2.2.100	51.104.174...	TCP	32786	80	http	GW-marke...	Not Suspic...		
Jul 22, 2026 4:42 AM	allow-internet...	2013504	Information	10.1.2.5	52.91.65.63	TCP	48652	80	http	GW-accou...	Not Suspic...		

DFW Engines At-a-Glance

- **eBPF** (extended Berkeley Packet Filter) Engine (L4) → Stateful Firewall Rule (forwarding path)
- WebProxy **ATS** (Apache Traffic Server) Engine (L7) → it is triggered whether WebGroups or TLS Decryption are required
- **Suricata** Engine (DPI) → Signature of the payload (only in IDS mode at the moment)





Next: Lab 11 – Distributed Cloud Firewall